Q2. An a.c. supply of 230 V is applied to a half-wave rectifier circuit through a transformer of turn ratio 10 : 1. Find (i) the output d.c. voltage and (ii) the peak inverse voltage. Assume the diode to be ideal.

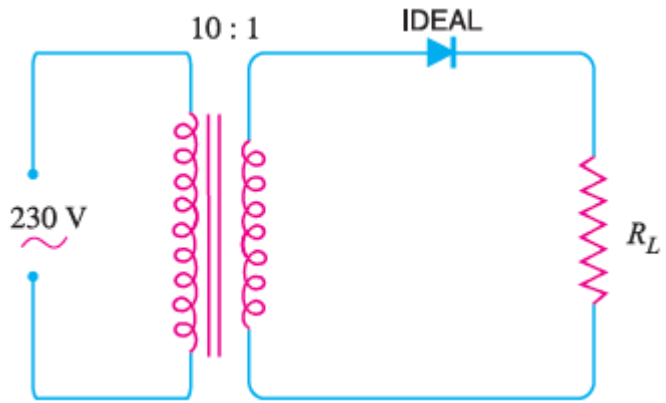


Fig. 1

Solution :

Primary to secondary turns is

$$\frac{N_1}{N_2} = 10$$

R.M.S. primary voltage

$= 230$ V

$\therefore$ Max. primary voltage is

$$V_{pm} = (\sqrt{2}) \times \text{r.m.s. primary voltage}$$

$$= (\sqrt{2}) \times 230 = 325.3 \text{ V}$$

Max. secondary voltage is

$$V_{sm} = V_{pm} \times \frac{N_2}{N_1} = 325.3 \times \frac{1}{10} = 32.53 \text{ V}$$

(i)

$$I_{d.c.} = \frac{I_m}{\pi}$$

$$V_{dc} = \frac{I_m}{\pi} \times R_L = \frac{V_{sm}}{\pi} = \frac{32.53}{\pi} = \mathbf{10.36\ V}$$

(ii) During the negative half-cycle of a.c. supply, the diode is reverse biased and hence conducts no current. Therefore, the maximum secondary voltage appears across the diode.

$$\therefore \quad \text{Peak inverse voltage} = \mathbf{32.53\ V}$$

Q3. A crystal diode having internal resistance rf = 20Ω is used for half-wave rectification. If the applied voltage v = 50 sin ω t and load resistance RL= 800 Ω, find :
(i) Im, Idc, Irms (ii) a.c. power input and d.c. power output (iii) d.c. output voltage (iv) efficiency of rectification.

Solution :

$$v = 50 \sin \omega t$$

$$\therefore \quad \text{Maximum voltage, } V_m = 50 \text{ V}$$

(i)

$$I_m = \frac{V_m}{r_f + R_L} = \frac{50}{20 + 800} = 0.061 \text{ A} = \mathbf{61 \text{ mA}}$$

$$I_{dc} = I_m/\pi = 61/\pi = \mathbf{19.4 \text{ mA}}$$

$$I_{rms} = I_m/2 = 61/2 = \mathbf{30.5 \text{ mA}}$$

(ii)

$$\text{a.c. power input} = (I_{rms})^2 \times (r_f + R_L) = \left(\frac{30.5}{1000}\right)^2 \times (20 + 800) = \mathbf{0.763 \text{ watt}}$$

$$\text{d.c. power output} = I_{dc}^2 \times R_L = \left(\frac{19.4}{1000}\right)^2 \times 800 = \mathbf{0.301 \text{ watt}}$$

(iii)

$$\text{d.c. output voltage} = I_{dc} R_L = 19.4 \text{ mA} \times 800 \, \Omega = \mathbf{15.52 \text{ volts}}$$

(iv)

$$\text{Efficiency of rectification} = \frac{0.301}{0.763} \times 100 = \mathbf{39.5\%}$$

Q4. A half-wave rectifier is used to supply 50V d.c. to a resistive load of 800 Ω. The diode has a resistance of 25 Ω. Calculate a.c. voltage required.

Solution :

Output d.c. voltage, $V_{dc} = 50$ V

Diode resistance, $r_f = 25\ \Omega$

Load resistance, $R_L = 800\ \Omega$

Let V_m be the maximum value of a.c. voltage required.

$$\therefore \quad V_{dc} = I_{dc} \times R_L$$

$$= \frac{I_m}{\pi} \times R_L = \frac{V_m}{\pi (r_f + R_L)} \times R_L \qquad \left[\because I_m = \frac{V_m}{r_f + R_L} \right]$$

$$\text{or} \quad 50 = \frac{V_m}{\pi (25 + 800)} \times 800$$

$$\therefore \quad V_m = \frac{\pi \times 825 \times 50}{800} = \mathbf{162\ V}$$

Hence a.c. voltage of maximum value 162 V is required.

Q5. A full-wave rectifier uses two diodes, the internal resistance of each diode may be assumed constant at 20 Ω. The transformer r.m.s. secondary voltage from centre tap to each end of secondary is 50 V and load resistance is 980 Ω. Find : (i) the mean load current (ii) the r.m.s. value of load current.

Solution :

$$r_f = 20\ \Omega, \quad R_L = 980\ \Omega$$

$$\text{Max. a.c. voltage, } V_m = 50 \times \sqrt{2} = 70.7\ \text{V}$$

$$\text{Max. load current, } I_m = \frac{V_m}{r_f + R_L} = \frac{70.7\ \text{V}}{(20 + 980)\ \Omega} = 70.7\ \text{mA}$$

(i)

$$\text{Mean load current, } I_{dc} = \frac{2I_m}{\pi} = \frac{2 \times 70.7}{\pi} = \mathbf{45\ mA}$$

(ii)

R.M.S. value of load current is

$$I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{70.7}{\sqrt{2}}\overline{AB} = \mathbf{50\ mA}$$

Q6. In the centre-tap circuit shown in Fig. 2, the diodes are assumed to be ideal i.e. having zero internal resistance. Find :(i) d.c. output voltage(ii) peak inverse voltage (iii) rectification efficiency.

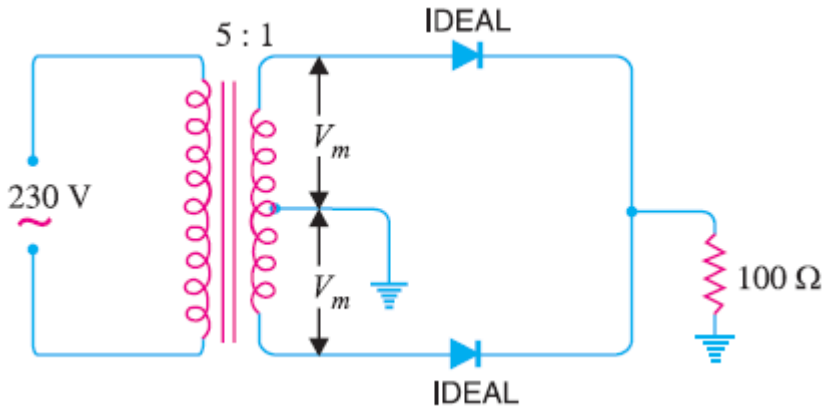


Fig. 2

Solution :

Primary to secondary turns, N1 / N2 = 5

R.M.S. primary voltage = 230 V

$\therefore$ R.M.S. secondary voltage

$= 230 \times (1/5) = 46$ V

Maximum voltage across secondary

$= 46 \times \sqrt{2} = 65$V

Maximum voltage across half secondary winding is

$V_m = 65/2 = 32.5$ V

(i) Average current, Idc =

$$\frac{2V_m}{\pi R_L} = \frac{2 \times 32.5}{\pi \times 100} = 0.207 \text{ A}$$

(ii) The peak inverse voltage is equal to the maximum secondary voltage, i.e

$$PIV = \mathbf{65\ V}$$

(iii)

$$\text{Rectification efficiency} = \frac{0.812}{1 + \frac{r_f}{R_L}}$$

Since $r_f = 0$

Rectification efficiency = **81.2 %**

Q7. In the bridge type circuit shown in Fig. 3, the diodes are assumed to be ideal. Find : (i) d.c. output voltage (ii) peak inverse voltage (iii) output frequency. Assume primary to secondary turns to be 4.

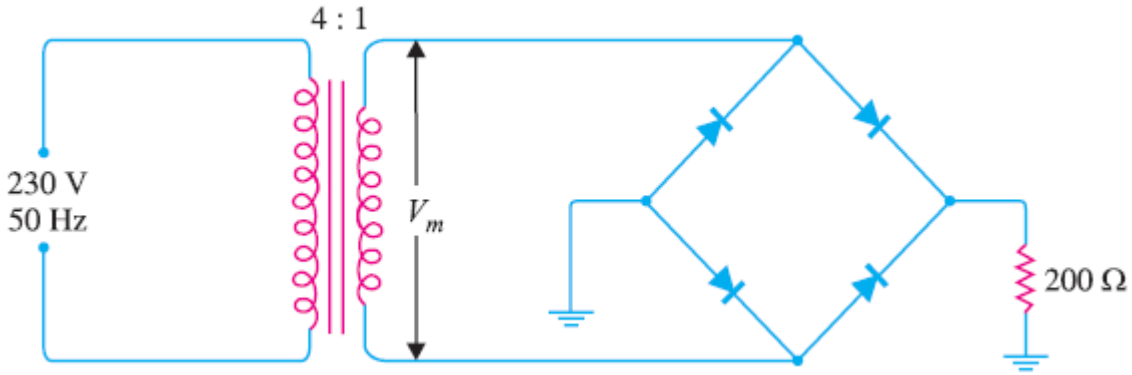


Fig.3

Solution :

Primary/secondary turns, $N_1/N_2 = 4$

R.M.S. primary voltage = 230 V

$\therefore$ R.M.S. secondary voltage = $230\ (N_2/N_1) = 230 \times (1/4) = 57.5$ V

Maximum voltage across secondary is

$$V_m = 57.5 \times \sqrt{2} = 81.3 \text{ V}$$

(*i*) Average current, $I_{dc} = \dfrac{2V_m}{\pi R_L} = \dfrac{2 \times 81.3}{\pi \times 200} = 0.26$ A

$\therefore$ d.c. output voltage, $V_{dc} = I_{dc} \times R_L = 0.26 \times 200 =$ **52 V**

(*ii*) The peak inverse voltage is equal to the maximum secondary voltage *i.e.*

$$PIV = 81.3 \text{ V}$$

(*iii*) In full-wave rectification, there are two output pulses for each complete cycle of the input a.c. voltage. Therefore, the output frequency is twice that of the a.c. supply frequency *i.e.*

$$f_{out} = 2 \times f_{in} = 2 \times 50 = 100 \text{ Hz}$$

Q8. Fig. 4(i) and Fig. 4 (ii) show the centre-tap and bridge type circuits having the same load resistance and transformer turn ratio. The primary of each is connected to 230V, 50 Hz supply. (i) Find the d.c. voltage in each case. (ii) PIV for each case for the same d.c. output. Assume the diodes to be ideal.

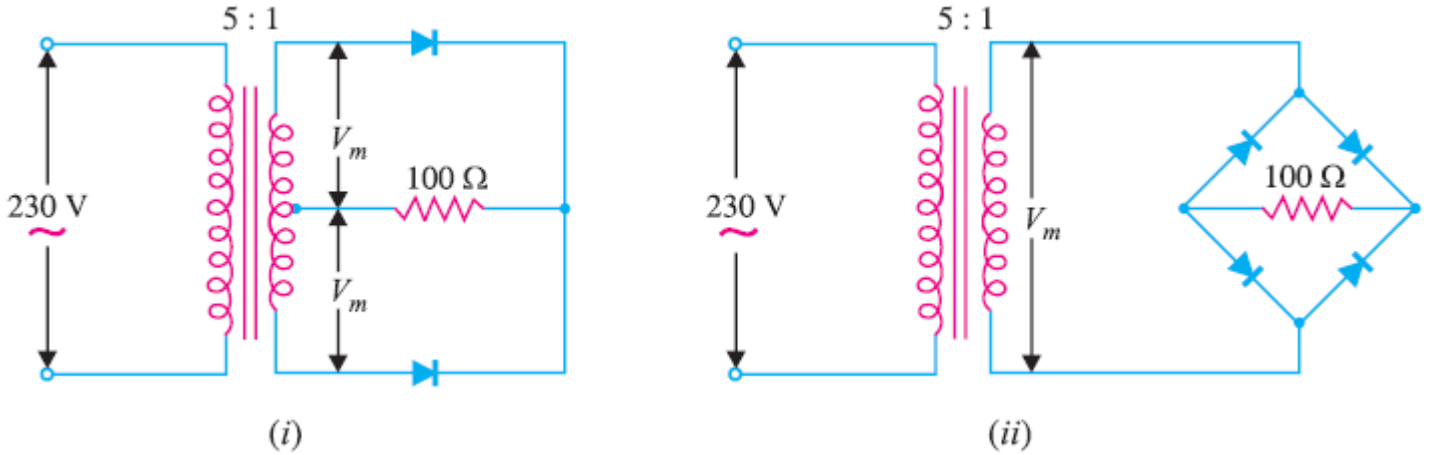


Fig. 4

Solution :

(i) DC output voltage :

Centre-tap circuit :

$$\text{R.M.S. secondary voltage} = 230 \times 1/5 = 46 \text{ V}$$

$$\text{Max. voltage across secondary} = 46 \times \sqrt{2} = 65 \text{ V}$$

Max. voltage appearing across half secondary winding is

$$V_m = 65/2 = 32.5 \text{ V}$$

$$\text{Average current, } I_{dc} = \frac{2V_m}{\pi R_L}$$

$$\text{D.C. output voltage, } V_{dc} = I_{dc} \times R_L = \frac{2V_m}{\pi R_L} \times R_L$$

$$= \frac{2V_m}{\pi} = \frac{2 \times 32.5}{\pi} = \mathbf{20.7 \text{ V}}$$

Bridge circuit :

$$\text{Max. voltage across secondary, } V_m = 65 \text{ V}$$

$$\text{D.C. output voltage, } V_{dc} = I_{dc} R_L = \frac{2V_m}{\pi R_L} \times R_L = \frac{2V_m}{\pi} = \frac{2 \times 65}{\pi} = \mathbf{41.4 \text{ V}}$$

This shows that for the same secondary voltage, the d.c. output voltage of bridge circuit is twice that of the centre-tap circuit

(ii) PIV for same d.c. output voltage :

The d.c. output voltage of the two circuits will be the same if Vm (i.e. max. voltage utilized by each circuit for conversion into d.c.) is the same. For this to happen, the turn ratio of the transformers should be as shown in Fig. 5.

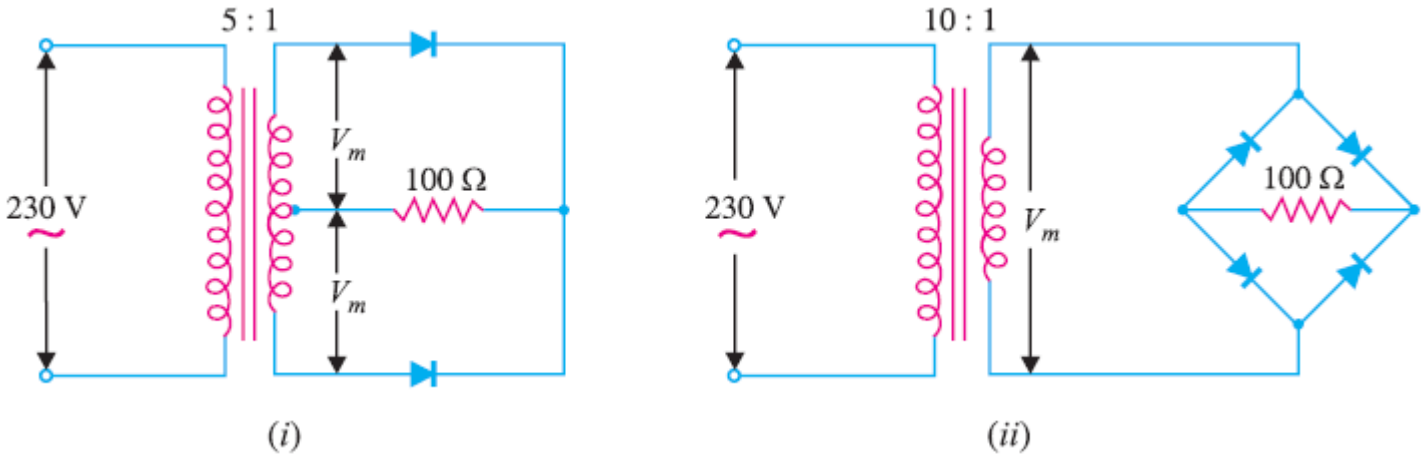


Fig.5

Centre-tap circuit :

$$\text{R.M.S. secondary voltage} = 230 \times 1/5 = 46 \text{ V}$$

$$\text{Max. voltage across secondary} = 46 \times \sqrt{2} = 65 \text{ V}$$

Max. voltage across half secondary winding is

$$V_m = 65/2 = 32.5 \text{ V}$$

$$\therefore \quad PIV = 2\,V_m = 2 \times 32.5 = \mathbf{65\ V}$$

Bridge circuit :

$$\text{R.M.S. secondary voltage} = 230 \times 1/10 = 23 \text{ V}$$

$$\text{Max. voltage across secondary, } V_m = 23 \times \sqrt{2} = 32.5 \text{ V}$$

$$\therefore \quad PIV = V_m = \mathbf{32.5\ V}$$

This shows that for the same d.c. output voltage, PIV of bridge circuit is half that of centre-tap circuit. This is a distinct advantage of bridge circuit.

Q9. The four diodes used in a bridge rectifier circuit have forward resistances which may be considered constant at 1Ω and infinite reverse resistance. The alternating supply voltage is 240 V r.m.s. and load resistance is 480 Ω. Calculate (i) mean load current and (ii) power dissipated in each diode.

Solution :

$$\text{Max. a.c. voltage, } V_m = 240 \times \sqrt{2} \text{ V}$$

(i) At any instant in the bridge rectifier, two diodes in series are conducting. Therefore, total circuit resistance = $2 r_f + R_L$.

$$\text{Max. load current, } I_m = \frac{V_m}{2 r_f + R_L} = \frac{240 \times \sqrt{2}}{2 \times 1 + 480} = 0.7 \text{ A}$$

$$\therefore \quad \text{Mean load current, } I_{dc} = \frac{2 I_m}{\pi} = \frac{2 \times 0.7}{\pi} = \mathbf{0.45 \text{ A}}$$

(ii) Since each diode conducts only half a cycle, diode r.m.s. current is :

$$I_{r.m.s.} = I_m/2 = 0.7/2 = 0.35 \text{ A}$$

$$\text{Power dissipated in each diode} = I^2_{r.m.s.} \times r_f = (0.35)^2 \times 1 = \mathbf{0.123 \text{ W}}$$

Q10. The bridge rectifier shown in Fig. 6 uses silicon diodes. Find (i) d.c. output voltage (ii) d.c. output current. Use simplified model for the diodes

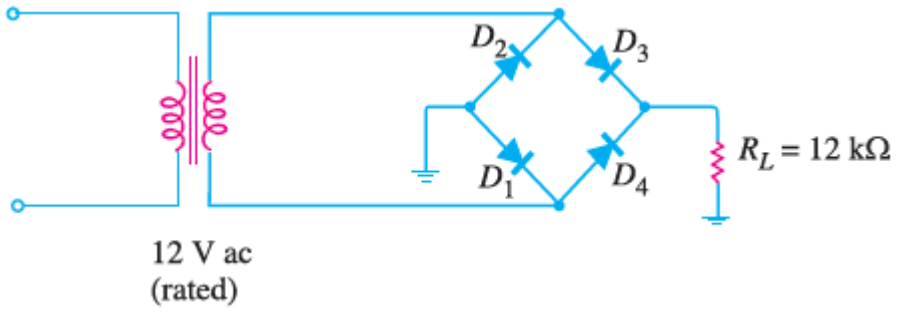


Fig.6

Solution :

The conditions of the problem suggest that the a.c voltage across transformer secondary is 12V r.m.s.

$\therefore$ Peak secondary voltage is

$$V_{s\,(pk)} = 12 \times \sqrt{2} = 16.97 \text{ V}$$

(i) At any instant in the bridge rectifier, two diodes in series are conducting.

$\therefore$ Peak output voltage is

$$V_{out\,(pk)} = 16.97 - 2\,(0.7) = 15.57 \text{ V}$$

$\therefore$ Average (or d.c.) output voltage is

$$V_{av} = V_{dc} = \frac{2\,V_{out\,(pk)}}{\pi} = \frac{2 \times 15.57}{\pi} = \mathbf{9.91V}$$

(ii) Average (or d.c.) output current is

$$I_{av} = \frac{V_{av}}{R_L} = \frac{9.91\text{V}}{12\ \text{k}\Omega} = \mathbf{825.8\ \mu A}$$

Q11. A power supply A delivers 10 V dc with a ripple of 0.5 V r.m.s. while the power supply B delivers 25 V dc with a ripple of 1 mV r.m.s. Which is better power supply ?

Solution :

The lower the ripple factor of a power supply, the better it is.

For power supply A

$$\text{Ripple factor} = \frac{V_{ac(r.m.s.)}}{V_{dc}} = \frac{0.5}{10} \times 100 = 5\%$$

For power supply B

$$\text{Ripple factor} = \frac{V_{ac(r.m.s.)}}{V_{dc}} = \frac{0.001}{25} \times 100 = 0.004\%$$

Clearly, power supply *B* is better.

Q12. For the circuit shown in Fig.7, find the output d.c. voltage.

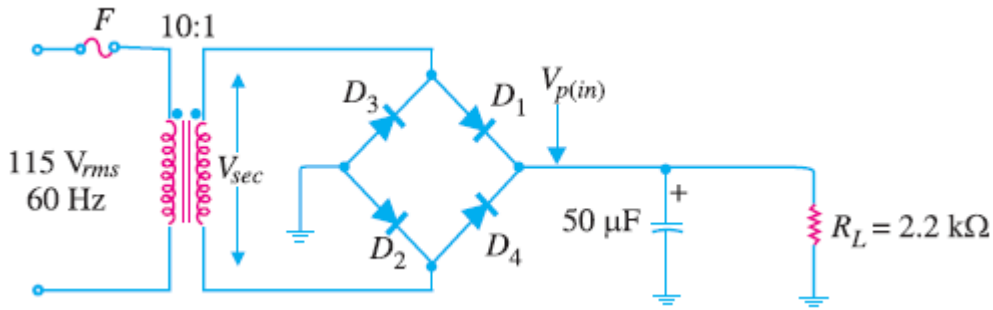


Fig.7

Solution :

It can be proved that output d.c. voltage is given by :

$$V_{dc} = V_{p(in)}\left(1 - \frac{1}{2f\ R_L C}\right)$$

Here $V_{p(in)}$ = Peak rectified full-wave voltage applied to the filter

f = Output frequency

Peak primary voltage, $V_{p(prim)} = \sqrt{2} \times 115 = 163\text{V}$

Peak secondary voltage, $V_{p(sec)} = \left(\frac{1}{10}\right) \times 163 = 16.3\text{V}$

Peak full-wave rectified voltage at the filter input is

$$V_{p\,(in)} = V_{p\,(sec)} - 2{\times}0.7 = 16.3 - 1.4 = 14.9\text{V}$$

For full-wave rectification, $f = 2\ f_{in} = 2{\times}60 = 120$ Hz

Now $$\frac{1}{2f\ R_L C} = \frac{1}{2 \times 120 \times (2.2{\times}10^{3}) \times (50 \times 10^{-6})} = 0.038$$

$$\therefore \quad V_{dc} = V_{p(in)}\left(1 - \frac{1}{2f\ R_L C}\right) = 14.9\,(1 - 0.038) = \mathbf{14.3V}$$

Q13. The choke of Fig.8 has a d.c. resistance of 25 Ω. What is the d.c. voltage if the full-wave signal into the choke has a peak value of 25.7 V ?

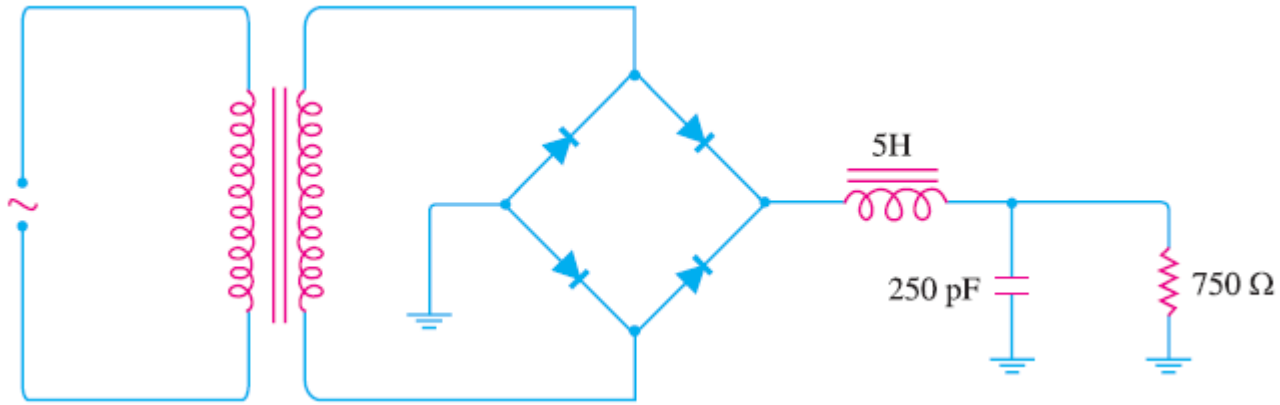


Fig.8

Solution :

The output of a full-wave rectifier has a d.c. component and an a.c. component. Due to the presence of a.c. component, the rectifier output has a pulsating character as shown in Fig. 9.

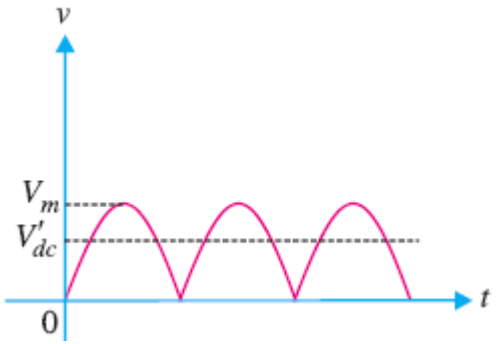


Fig. 9

The maximum value of the pulsating output is Vm and d.c. component is V′dc = 2 Vm/π.

For d.c. component V′dc, the choke resistance is in series with the load as shown in 10.

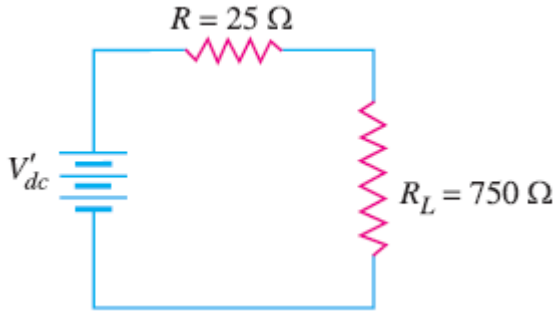


Fig. 10

$$\therefore \quad \text{Voltage across load, } V_{dc} = \frac{V'_{dc}}{R + R_L} \times R_L$$

$$\text{In our example, } V'_{dc} = \frac{2V_m}{\pi} = \frac{2 \times 25.7}{\pi} = 16.4 \text{ V}$$

$$\therefore \quad \text{Voltage across load, } V_{dc} = \frac{V'_{dc}}{R + R_L} \times R_L = \frac{16.4}{25 + 750} \times 750 = \mathbf{15.9\ V}$$

The voltage across the load is 15.9 V dc *plus* a small ripple.